Replication Exercise: "Improving the Design of Conditional Transfer Programs: Evidence from a Randomized Education Experiment in Colombia"*

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Introduction

Barrera-Osorio et al. (2011) conduct an experimental evaluation of different conditional cash transfer (CCT) programs designed to incentivize academic participation in Colombia. In particular, the authors compare three different treatment branches: a standard (or "basic" treatment) design similarly to the well-known PROGRESA[1] program, a "savings" treatment wherein one-third of funds are distributed in a lump-sum prior to students' re-enrollment dates, and a "tertiary" treatment which guarantees a large payment upon graduation, directly incentivizing secondary school graduation and enrollment in tertiary institutions.

In this paper, we will replicate the results from Barrera-Osorio et al. (2011). We have chosen to study this experiment in particular for multiple reasons. At a high level, we are strongly interested in education and development, especially in the context of Latin America. Moreover, with respect to the aim of this course, we believe this experiment contains several desirable characteristics. Notably, the presence of three different treatments arms allows us to explore heterogeneity in multiple settings with different motivations, resulting in more meaningful policy conclusions. In addition, the availability of monitored as well as self-reported outcome data allows us to identify systematic biases in survey answers which could obscure heterogeneity in treatment effects. Finally, the authors' own heterogeneity analysis is rather limited, with parametric assumptions naturally restricting and potentially biasing our understanding. With this in mind, we intend to use machine learning methods to explore and uncover treatment effect heterogeneity in an unrestricted and systematic fashion (*principled* data snooping, if you will).

Methodological Setting

Randomizing treatment at the student-level (generating variation within schools and also within families) and stratifying based on locality, public/private school, gender, and grade level for the 13,433 children registered, the authors control for confounding factors which could otherwise bias causal estimates of the treatment effects.[2]

*Barrera-Osorio et al. (2011)

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[1]See Gertler (2004).

[2]More formally, the randomized controlled trial (RCT) framework ensures ignorability, i.e. $Y_{(1)}, Y_{(0)} \perp D$ for Y_i potential outcomes and D treatment status, and therefore associations found can be interpreted causally.

We will depart from their following baseline specifications:[3]

$$y_{ij} = \beta_0 + \beta_B Basic_i + \beta_S Savings_i + \varepsilon_{ij} \tag{1}$$

$$y_{ij} = \beta_0 + \beta_B Basic_i + \beta_S Savings_i + \delta X_{ijk} + \phi_j + \varepsilon_{ij} \tag{2}$$

$$y_{ij} = \beta_0 + \beta_B Basic_i + \beta_S Savings_i + \sum_{m=1}^{M} \alpha_{0m} X_{ijk}^m + \sum_{m=1}^{M} (\alpha_{1m} Basic_i + \alpha_{2m} Savings_i) X_{ijk}^m + \phi_j + \varepsilon_{ij} \tag{3}$$

Given (proper) randomization, Equation 1 consistently identifies β_B and β_S, the coefficients for indicator variables of belonging to each treatment status, using ordinary least squares (OLS). Equation 2 introduces controls X_{ijk} and school-level fixed effects ϕ_j to account for imperfect randomization (and potentially reduce variance). Equation 3 introduces interaction terms $\forall m \in M, \;\; D_i * X_{ijk}^m$, such that X_{ijk} is a $n \;\; x \;\; M$ matrix composed of every X_{ijk}^m, to the right-hand-side of Equation 2 to explore treatment effect heterogeneity.

Figure 1 presents the original results of Equation 1 and Equation 2 and Table 1 gathers our replication, while Figure 2 depicts the estimates of Equation 3 and Table 2 our replication, respectively. Barrera-Osorio et al. (2011) present a rather limited heterogeneity analysis, based on a linear regression model interacting the first two treatment branches (standard and savings) with gender, income-poor and a continuous predictor of enrollment and attendance absent treatment. Their results shed light on policy-relevant considerations related to program targeting and have a meaningful economic interpretation in the specific framework of educational CCTs. Nevertheless, the authors plausibly neglect other important sources of heterogeneity. In the next section, then, we explain how to complement and extend their analysis using machine learning algorithms.

Exploring Treatment Effect Heterogeneity

In certain contexts, researchers may be concerned with heterogeneous treatment effects across only certain, known characteristics. In many other contexts, however, it may not be known *a priori* which dimensions may exhibit statistically (as well as potentially economically) significant variation. Further, it may well be infeasible to conduct comprehensive heterogeneity analysis on a case-by-case basis, a strategy which would, in any case, be prone to multiple hypothesis testing errors. Machine learning algorithms can, in such situations, provide a systematic way to investigate heterogeneity. With that motivation in mind, the aim of this section is to firstly introduce the machine learning algorithms used, secondly explain the hyperparameter selection process for a subset of them, and finally comment briefly on the results and consequent economic interpretations.

We start by employing all the tools available to uncover treatment effect heterogeneity. Notably, we use OLS with interactions, post-lasso selection, (honest) causal trees (HCT) and forests (HCF), best linear predictor (BLP) and group sorted average treatment effects (GATES). Implementing and comparing these models allows us insight not only into the sources of heterogeneity found in each, but also into the relative volatility and consistency across algorithms.

Figure 3 and Figure 4 respectively provide, for the standard and savings treatment arms, a graphical comparison of the distributions of the *conditional* average treatment effects (CATEs) on attendance identified in several models. Some methods, like OLS and causal forest, generate stable, (quasi-) normal distributions. Unsurprisingly, vis-a-vis the causal forest, the causal tree (Figure 5) shows more volatility. Post-selection lasso is strongly dependent on the penalty hyperparameter λ, chosen using 5-fold cross-validation minimizing the

[3] The same functional specifications work for the so-called tertiary treatment.

(validation) mean squared error (MSE). Based on how well a given set of covariates and interactions predict the outcome variable *and* the treatment status, heterogeneity present as a function of weak predictors may be neglected.

In Table 3, treatment effect magnitudes and standard errors are explored. The lower half of coefficients correspond to the the interaction terms between leaves and treatment. We see that only one of the six leaves, "leaf0.0448" has a significant treatment effect when interacted with treatment. And ultimately, there are no significant differences (to the 5% level) in treatment effects between groups (here, the leaves shown in Figure 5).

This constitutes a preliminary approximation of the methods' relative performances and characteristics, as each methodology relies on a different set of assumptions and algorithmic procedures to fine-tune parameters. Nonetheless, the desirability of each procedure is strongly context-specific.

In a more general sense, BLP allows us to test an important result. Rejecting the null $\beta_2 = 0$ in

$$Y_{ijk} = \alpha X_{ijk} + \beta_1(Basic_i - p(X_{ijk})) + \beta_2(Basic_i - p(X_{ijk}))(\hat{\tau}(X_{ijk}) - \mathbb{E}[\hat{\tau}(X_{ijk})]) + \upsilon_{ijk} \tag{4}$$

would imply that the CATE estimates are well designed, and that there is some degree of (unidentified) treatment effect heterogeneity. Nonetheless, we are unable to reject it for both treatment arms, and thus we cannot disentangle whether there is no sufficient variation in CATEs or whether our CATE estimates are incorrectly specified. In this particular case, Table 4 and Table 5 gather the relevant information. On the other side, BLP is inherently complex to interpret, as the source of heterogeneity cannot be identified. Extending the analysis employing GATES could solve this concern applying classification analysis (CLAN) to characterize the generated groups based on observables. GATES results are presented in Table 6, with no significant effect detected.

Conclusions

In this paper we have conducted a re-evaluation of the multiple-arms experiment studies in Barrera-Osorio et al. (2011), aimed at sheding light at treatment effect heterogeneity using machine learning techniques.

Our main conclusions can be subsumed in three points. Firstly, their heterogeneity analysis is inherently limited in nature, as it may suffer from functional misspecification and theoretical biases. Secondly, machine learning algorithms may be of much use to identify treatment effect heterogeneity under ignorability, without imposing any *ex ante* restriction or without relying on economic theory. They are, nevertheless, operating at different levels, implying that their desirability is highly context-specific and that they will uncover heterogeneity in very different ways. For instance, Barrera-Osorio et al. (2011) consider *a priori* that long-term credit-constrained families may strongly benefit from the savings treatment, and include an interaction with a "poor" dummy as a result. OLS and post-selection lasso may capture this as well, but HCT, HCF or GATES are likely to pick up non-linearities in a less interpretable way, meaning they may split income differently and may yield direct comparison complicated. Finally, the presence of several treatment arms allows us to evaluate whether different policy schedules should be targeted differently. Machine learning mechanisms and the original results do not categorically point in this direction (checked through multiple hypothesis testing).

References

Barrera-Osorio, F., Bertrand, M., Linden, L. L., and Perez-Calle, F. (2011). Improving the design of conditional transfer programs: Evidence from a randomized education experiment in colombia. *American Economic Journal: Applied Economics*, 3(2):167–95.

Gertler, P. (2004). Do conditional cash transfers improve child health? evidence from progresa's control randomized experiment. *American Economic Review*, 94(2):336–341.

Appendix

Figure 1: Effects on Monitored School Attendance Rates (Table 3 Original Document)

	Basic-Savings			Tertiary			Both
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Basic treatment	0.033***	0.032***	0.032***				0.025**
	(0.007)	(0.008)	(0.007)				(0.010)
Savings treatment	0.029***	0.027***	0.027***				0.028**
	(0.008)	(0.008)	(0.007)				(0.012)
Tertiary treatment				0.052***	0.054***	0.056**	0.055***
				(0.018)	(0.016)	(0.020)	(0.020)
H0: Basic—Savings							
F-Stat	0.31	0.4	0.48				0.05
p-value	0.58	0.53	0.49				0.82
H0: Tertiary—Basic							
F-Stat							1.87
p-value							0.18
Demographic controls		√	√		√	√	√
School fixed effects			√			√	√
Observations	5,799	5,799	5,799	930	930	930	2,937
R^2	< 0.01	0.04	0.09	0.01	0.06	0.27	0.13
Control average	0.792	0.792	0.792	0.776	0.776	0.776	0.786

This table displays the estimated effects of the respective treatments on students' monitored attendance rates. Columns 1, 2, and 3 contain estimates of the effects of the basic and savings treatments in San Cristobal in grades 6–10. Columns 4, 5, and 6 contain estimates of the treatment effect for the tertiary treatment in Suba for grades 9 and 10. Column 7 contains estimates of the effects of all three treatments on students in grades 9 and 10 for all localities. Control variables include a rich set of socio-demographic variables (indicators for the households' ownership of assets, access to utilities, possession of durable goods, and physical infrastructure of the house), the square of the child's age, the number of years too old a child is for his or her grade, and indicator variables for the child's marital status, the family's marital status, grade, and whether or not the child is over the average age for his or her grade. Standard errors are clustered at the school level. Significant at the (*** 1% level, ** 5% level, * 10% level).

Table 1: Effects on Monitored School Attendance Rates (Replication)

	Basic-Savings			Tertiary			Both
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Basic Treatment	0.033***	0.032***	0.032***				0.024**
	(0.007)	(0.008)	(0.007)				(0.010)
Savings Treatment	0.029***	0.028***	0.029***				0.030***
	(0.008)	(0.008)	(0.008)				(0.011)
Tertiary Treatment				0.052***	0.054***	0.054***	0.054***
				(0.018)	(0.017)	(0.017)	(0.017)
Demographic Controls		✓	✓		✓	✓	✓
School Fixed Effects			✓			✓	✓
Observations	5,799	5,799	5,799	930	930	930	2,937
R^2	0.003	0.029	0.030	0.008	0.048	0.051	0.024
Adjusted R^2	0.003	0.026	0.026	0.007	0.028	0.030	0.016

This table replicates Figure 1 (Table 3 in Barrera-Osorio et al., 2011), employing the same subsets of controls they indicate in the publication and in their Stata replication files, school fixed effects and clustering standard errors at the school level. Some slight differences may be the result of choosing different reference categories for categorical variables, or different cluster-robust formulae for standard errors. Nonetheless, this replication exercise ensures we are on the right track, understanding both their functional specifications and the underlying data-generating process for the application of ML algorithms. Significant at the (*** 1% level, ** 5% level, * 10% level).

Figure 2: Variation in the Basic and Savings Treatment Effects, Basic-Savings Experiment (Table 5 Original Document)

	Gender		Income		Projected participation	
	Attendance	Enrollment	Attendance	Enrollment	Attendance	Enrollment
	(1)	(2)	(3)	(4)	(5)	(6)
Basic treatment	0.049***	0.016	0.067***	0.031*	0.225***	0.03
	(0.012)	(0.014)	(0.019)	(0.018)	(0.078)	(0.066)
Savings treatment	0.050***	0.048***	0.047***	0.043**	0.165**	0.153***
	(0.010)	(0.015)	(0.013)	(0.017)	(0.066)	(0.055)
Basic × Girl	−0.034*	−0.011				
	(0.019)	(0.019)				
Savings × Girl	−0.045***	−0.016				
	(0.013)	(0.018)				
Girl	0.032***	0.025				
	(0.011)	(0.015)				
Basic × low income			−0.050**	−0.029		
			(0.021)	(0.023)		
Savings × low income			−0.029**	−0.006		
			(0.013)	(0.021)		
Low income			0.014	0.007		
			(0.015)	(0.018)		
Basic × projected enrollment						−0.027
						(0.085)
Savings × projected enrollment						−0.161**
						(0.074)
Basic × projected attendance					−0.244**	
					(0.094)	
Savings × projected attendance					−0.174**	
					(0.078)	
Demographic controls	√	√	√	√	√	√
School fixed effects	√	√	√	√	√	√
Observations	5,799	8,980	5,799	8,980	5,799	8,980
R^2	0.09	0.17	0.09	0.17	0.09	0.17

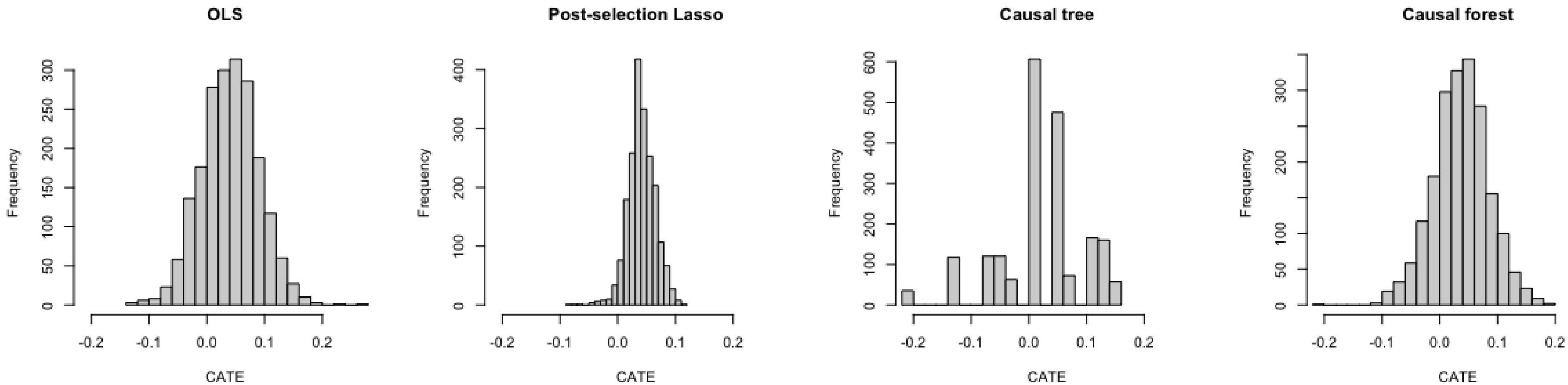
This table displays the results on variation in the treatment effects for the basic and savings treatments from Tables 3 and 4. Columns 1 and 2 contain results from interactions with the child's gender. Columns 3 and 4 contain results from interactions with income ("low income" is defined as a family earning less than 380,000 pesos or US $190 a month, this is the sixty-sixth percentile for income). Columns 5 and 6 contain interactions with the pro- jected attendance and enrollment variables. Note that all estimations include the same set of control variables, but these variables are only reported if they are interacted with the treatment variables to make the table clearer. Control variables include a rich set of sociodemographic variables (indicators for the households' ownership of assets, access to utilities, possession of durable goods, and physical infrastructure of the house), the square of the child's age, the number of years too old a child is for his or her grade, and indicator variables for the child's marital status, the family's marital status, grade, and whether or not the child is over the average age for his or her grade. Standard errors are clustered at the school level. Significant at the (*** 1% level, ** 5% level, * 10% level).

Table 2: Treatment Effect Heterogeneity (Replication)

	Gender		Income		Projected Participation	
	Attendance	Enrollment	Attendance	Enrollment	Attendance	Enrollment
	(1)	(2)	(3)	(4)	(5)	(6)
Basic Treatment	0.047***	0.016***	0.032***	0.030*	0.225**	0.03
	(0.012)	(0.014)	(0.007)	(0.017)	(0.083)	(0.041)
Savings Treatment	0.049***	0.048**	0.029***	0.044*	0.164**	0.153***
	(0.012)	(0.015)	(0.015)	(0.019)	(0.087)	(0.041)
Basic x Girl	−0.033*	−0.011				
	(0.017)	(0.016)				
Savings x Girl	−0.044**	−0.016				
	(0.016)	(0.021)				
Girl	0.032***	0.025				
	(0.011)	(0.015)				
Basic x low income			−0.051**	−0.028		
			(0.019)	(0.023)		
Savings x low income			−0.028	−0.006		
			(0.018)	(0.022)		
Low income			0.014	0.008		
			(0.015)	(0.019)		
Basic x Proj. Attendance					−0.243**	
					(0.094)	
Savings x Proj. Attendance					−0.174*	
					(0.0104)	
Basic x Proj. Enrollment						−0.028
						(0.057)
Savings x Proj. Enrollment						−0.161**
						(0.057)
Demographic Controls	✓	✓	✓	✓	✓	✓
School Fixed Effects	✓	✓	✓	✓	✓	✓
Observations	5,799	8,980	5,799	8,980	5,799	8,980
R^2	0.083	0.156	0.083	0.0156	0.083	0.157

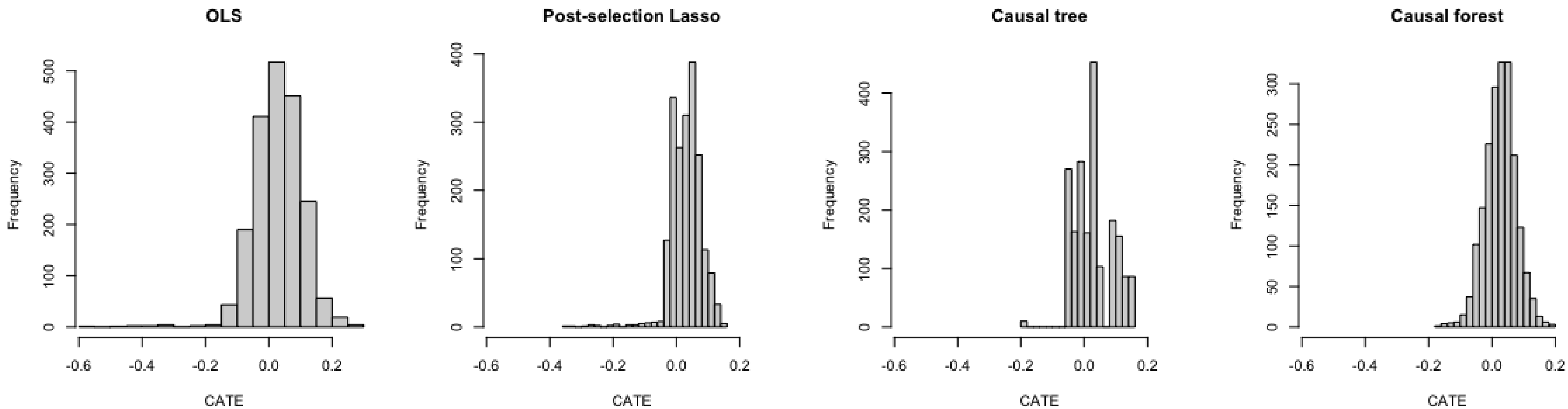
This table replicates Figure 2 (Table 5 in Barrera-Osorio et al., 2011), employing the same subsets of controls they indicate in the publication and in their Stata replication files, school fixed effects and clustering standard errors at the school level. Some slight differences may be the result of choosing different reference categories for categorical variables, or different cluster-robust formulae for standard errors. Nonetheless, this replication exercise ensures we are on the right track, understanding both their functional specifications and the underlying data-generating process for the application of ML algorithms. Significant at the (*** 1% level, ** 5% level, * 10% level).

Figure 3: Distribution of standard treatment CATE on monitored attendance across methods



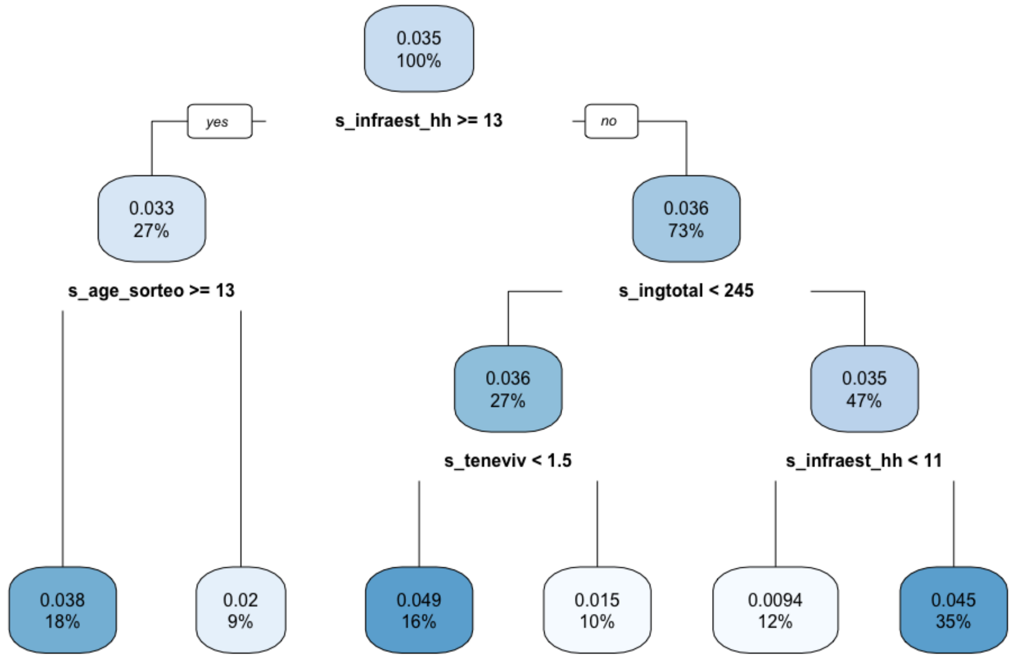
This figure provides some preliminary insights on the CATEs generated by each method under a basic set-up with arbitrarily chosen hyperparameters. Post-selection Lasso generates the least variation; the causal tree depicts an unstable distribution (result of its high variance) and the OLS and the causal forest result in a somehow symmetric distribution, centered in the ATE, which is (given the support of the response variable) significantly different from zero but close to it. CATEs for BLP and GATES are strongly dependent on the fitted model they rely on (which, in our case, is chosen through 5-fold cross-validation based on the power to predict Y), and so are not reported as a result.

Figure 4: Distribution of savings treatment CATE on monitored attendance across methods



This figure provides some preliminary insights on the CATEs generated by each method under a basic set-up with arbitrarily chosen hyperparameters. Post-selection Lasso generates the least variation; the causal tree depicts an unstable distribution (result of its high variance) and the OLS and the causal forest result in a somehow symmetric distribution, centered in the ATE, which is (given the support of the response variable) significantly different from zero but close to it. CATEs for BLP and GATES are strongly dependent on the fitted model they rely on (which, in our case, is chosen through 5-fold cross-validation based on the power to predict Y), and so are not reported as a result.

Figure 5: Honest Causal Tree



This visualization presents the highly accessible, though volatile honest causal tree trained on (half of) our data. We have specified that each leaf contain over one-twentieth of the treated as well as control cases. What it lacks in consistency, the tree may make up for in the interpretability of each (greedy) split. Here, for example, we learn that splitting on household income (*s_ingtotal*), age (*s_age_sorteo*), and infrastructure of house (*s_infraest_hh*) capture the greatest variation, followed by splitting on the house possession type (encoded as ordered variable *s_teneviv*) and again a different infrastructure split. From a research perspective, these covariates (along with other correlated variables) are worth investigating, especially if policymakers could address or exploit the variation in these features. Worth noting again, of course, is that changing the hyperparameters of the algorithm (i.e. the pruning level) as well as the data trained on (i.e. the randomization in the training versus validation splits) leads to different possible conclusions.

Table 3: Estimating the Treatment Effect Magnitudes and Standard Errors

	Dependent variable:
	Attendance
leaf0.0094	0.778***
	(0.025)
leaf0.0152	0.808***
	(0.028)
leaf0.0196	0.818***
	(0.030)
leaf0.0384	0.789***
	(0.020)
leaf0.0448	0.787***
	(0.014)
leaf0.0489	0.785***
	(0.021)
leaf0.0094:X	0.009
	(0.036)
leaf0.0152:X	0.015
	(0.038)
leaf0.0196:X	0.020
	(0.042)
leaf0.0384:X	0.038
	(0.029)
leaf0.0448:X	0.045**
	(0.021)
leaf0.0489:X	0.049
	(0.031)
Observations	1,996
R^2	0.896
Adjusted R^2	0.896
Residual Std. Error	0.275 (df = 1984)
F Statistic	1,428.891*** (df = 12; 1984)

*p<0.1; **p<0.05; ***p<0.01

This table displays the estimated treatment effects for each "group", where each group corresponds to the leaves of the HCT in Figure 5. Leaf0.0094 corresponds to the fifth leaf from the left in Figure 5, leaf0.0152 the fourth from the left, leaf0.0196 the second, and so forth. The lower half, of course, represents the coefficients associated with the interactions between said leaves and treatment (X). Notably, only leaf0.0448:X, i.e. the interaction between treatment and a infrastructure of the house below the mean, is significant to the 5% level. Nevertheless, it is important to note that despite only one group showing a significant treatment effect, there could still exist significant *difference* in treatment effects *between* groups. However, in this case, we do not find differences significantly different from zero at the 5% level, indicating a lack of treatment effect heterogeneity.

Table 4: BLP estimates: attendance under the standard treatment

Coefficient	**Estimate**	**Confidence interval** (90%)	
		Lower bound	**Upper bound**
$\hat{\beta}_1$	0.0232	0.000098	0.047
$\hat{\beta}_2$	-0.0967	-0.573	0.380

Table 5: BLP estimates: attendance under the savings treatment

Coefficient	**Estimate**	**Confidence interval** (90%)	
		Lower bound	**Upper bound**
$\hat{\beta}_1$	0.0282	0.006	0.051
$\hat{\beta}_2$	-0.416	-1.03	0.198

Table 6: GATES estimates: attendance under the standard treatment for Q = 10

Coefficient	**Estimate**	**Confidence interval** (90%)	
		Lower bound	**Upper bound**
γ_1	0.0715	-0.000475	0.144
γ_2	-0.0284	-0.103	0.0462
γ_3	0.0225	-0.0534	0.0983
γ_4	0.00945	-0.0656	0.0844
γ_5	0.0655	-0.00756	0.139
γ_6	-0.0622	-0.137	0.0123
γ_7	0.0664	-0.00718	0.140
γ_8	0.0398	-0.0345	0.114
γ_9	0.0399	-0.0340	0.114
γ_{10}	0.00378	-0.0695	0.0770